CO phase selection for a stirred catalytic surface model

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15 October 2026

Abstract

We consider a catalytic surface model on $\mathbb{Z}^2$ with CO adsorption, diatomic oxygen adsorption, finite-rate reaction and finite-rate stirring. If the CO arrival rate p exceeds the oxygen parameter q, we prove that every site is eventually occupied by CO almost surely, from every iid initial law with positive CO density. This gives CO poisoning on every finite set. The density estimate of Durrett and Swindle controls the total time spent vacant. An additional reaction count then controls all state changes, including stirring. Spatial independence of the initial states and driving clocks determines the limiting phase.

1 The model and the result

In the catalytic oxidation model of Durrett and Swindle [1], a surface site is vacant, occupied by a CO molecule, or occupied by an oxygen atom. Neighboring CO and oxygen react and leave two vacant sites. The two configurations consisting entirely of CO or entirely of oxygen are absorbing; either one stops the oxidation reaction.

Write $\eta_t(x) \in \{0, 1, 2\}$, $x \in \mathbb{Z}^2$, for the state at time t, with 0 vacant, 1 CO and 2 oxygen. Sites x, y are neighbors if $|x - y|_1 = 1$. The transition rates are

location	update	rate
site	$0 \longrightarrow 1$	p
unordered neighboring pair	$(0,0) \longrightarrow (2,2)$	$q/4$
unordered neighboring pair	$(1,2), (2,1) \longrightarrow (0,0)$	$r/4$
unordered neighboring pair	$(a,b) \longrightarrow (b,a)$	γ.

In the last row $a, b \in \{0, 1, 2\}$; exchanging equal states has no effect. All rates below are finite and satisfy

$$p > q > 0, \qquad r > 0, \qquad \gamma > 0.$$

Each update has its own independent Poisson clock. The initial states are independent of these clocks. Reaction is at a finite rate, rather than instantaneous.

Durrett and Swindle [1, Theorem 1 and its proof, pp. 490–492] showed that the vacant-site and neighboring mixed-species probabilities vanish when $p > q$, under translation-invariant initial laws. Thus every fixed finite box becomes monochromatic with probability tending to one. Selecting the CO phase requires a further argument. It is posed in [2, Section 5.4, Open Problem 5.4.6] for more general initial configurations. Here we prove selection for iid initial laws of positive CO density.

Theorem 1. *For the process above, let the initial states be iid with $\mathbb{P}(\eta_0(0) = 1) > 0$. Almost surely, for every $x \in \mathbb{Z}^2$, $\eta_t(x) = 1$ for all sufficiently large t. Consequently, for every finite $K \subset \mathbb{Z}^2$,*

$$\lim_{t\to\infty} \mathbb{P}(\eta_t(x) = 1 \text{ for every } x \in K) = 1. \tag{1}$$

The time after which a site stays in state 1 may depend on the site. The theorem does not assert that the whole infinite lattice is absorbed at one finite time. Its conclusion holds for each fixed admissible parameter choice, without a uniform convergence rate.

The proof retains the signed-density estimate from [1] and uses the CO balance once more to bound the total number of reactions. These two bounds control the expected number of each kind of state change. Once all sites fix, connectedness leaves only the two constant configurations, and the iid driving data make the choice of color deterministic. The signed density then excludes oxygen.

2 There are finitely many state changes

We use the graphical process driven by the independent clocks described above. It is a measurable, nonanticipating function of the initial states and clocks and commutes with spatial translations. If C is any driving clock of rate c, $F \subset \mathbb{Z}^2$ is finite and h is a bounded function of the states in F, then

$$\mathbb{E} \sum_{s \in C \cap (0,T]} h(\eta_{s-}|_F) = c \int_0^T \mathbb{E} h(\eta_t|_F)\, dt. \tag{2}$$

Appendix A gives the construction and a direct proof of this counting identity. It applies both to state increments and to the indicator that a clock ring changes a given site.

Let $a_i = \mathbb{P}(\eta_0(0) = i)$ and $u_i(t) = \mathbb{P}(\eta_t(0) = i)$, $i = 0, 1, 2$. To count oxygen additions and CO losses at the origin, put

$$O(t) = \frac{q}{4} \sum_{y \sim 0} \mathbb{P}(\eta_t(0) = 0, \eta_t(y) = 0),$$
$$R(t) = \frac{r}{4} \sum_{y \sim 0} \mathbb{P}(\eta_t(0) = 1, \eta_t(y) = 2).$$

Every sum here has four terms. Translation invariance, with y replaced by $-y$, gives

$$\sum_{y \sim 0} \mathbb{P}(\eta_t(0) = 2, \eta_t(y) = 1) = \sum_{y \sim 0} \mathbb{P}(\eta_t(0) = 1, \eta_t(y) = 2). \tag{3}$$

Stirring contributes zero to a one-site density, since its contribution for state i is $\gamma \sum_{y \sim 0} [u_i(t) - u_i(t)]$. Each effective oxygen ring on an incident edge adds one oxygen atom at the origin. Applying (2) to the state indicators therefore gives the exact balances

$$u_1(T) - a_1 = \int_0^T [p u_0(t) - R(t)]\, dt, \tag{4}$$
$$u_2(T) - a_2 = \int_0^T [O(t) - R(t)]\, dt. \tag{5}$$

No independence of neighboring states at positive times is used.

Set $m(t) = u_1(t) - u_2(t)$. Since $O(t) \le q u_0(t)$, subtraction yields

$$m(T) - m(0) = \int_0^T [p u_0(t) - O(t)]\, dt \ge (p - q) \int_0^T u_0(t)\, dt \ge 0. \tag{6}$$

This is the density argument in [1]. The bound $m(T) \le 1$ implies

$$I := \int_0^\infty u_0(t)\, dt \le \frac{1 - m(0)}{p - q} < \infty. \tag{7}$$

The CO balance (4) also gives

$$J := \int_0^\infty R(t)\,dt \le a_1 + pI < \infty, \tag{8}$$

because $u_1(T) \ge 0$. These limits use monotone convergence of nonnegative integrals.

Lemma 2. *Under the hypotheses of Theorem 1, almost surely every site changes state only finitely many times.*

Proof. Let $N_0(T)$ count state changes at the origin up to time T. The expected numbers due to CO arrivals, oxygen arrivals and reactions are, respectively,

$$p\int_0^T u_0(t)\,dt \le pI, \qquad \int_0^T O(t)\,dt \le qI, \qquad 2\int_0^T R(t)\,dt \le 2J.$$

The factor 2 in the reaction count comes from the two possible colors at the origin in (3).

For stirring let $D_y(t) = \mathbb{P}(\eta_t(0) \ne \eta_t(y))$. An unequal pair either contains a vacancy or consists of a CO and an oxygen atom, so

$$D_y(t) \le 2u_0(t) + \mathbb{P}(\eta_t(0) = 1, \eta_t(y) = 2) + \mathbb{P}(\eta_t(0) = 2, \eta_t(y) = 1).$$

Thus (3) bounds the expected number of effective exchanges at the origin by

$$\gamma\int_0^T \sum_{y\sim 0} D_y(t)\,dt \le 8\gamma I + \frac{8\gamma}{r}J.$$

Every change has been counted. Letting T increase to infinity,

$$\mathbb{E}N_0(\infty) \le (p + q + 8\gamma)I + (2 + 8\gamma/r)J < \infty. \tag{9}$$

Hence $N_0(\infty)$ is finite almost surely. Translation invariance gives the same conclusion at each site; the countable union of the exceptional null events still has probability zero. □

3 Selection of the limiting color

Lemma 3. *Almost surely the eventual states of all sites are the same and belong to* $\{1, 2\}$.

Proof. By Lemma 2, every site has an eventual state. Simultaneously for all the countably many positive-rate clocks, their ring times are unbounded almost surely. Indeed, for a rate-c clock the probability of no ring after an integer n is $\lim_{M\to\infty} e^{-c(M-n)} = 0$.

An eventual vacancy at a site would be changed by a later CO ring. Two neighboring eventual states 1 and 2 would be changed by a reaction ring after both sites had fixed. Neither is possible. Connectedness of $\mathbb{Z}^2$ now makes the eventual configuration constant. These are sample-path arguments and require no stopping-time property of the last-change times. □

Proof of Theorem 1. Let e_1, e_2 be the coordinate unit vectors. At each x, group its initial state and site clock with all three clocks on each of $\{x, x + e_1\}$ and $\{x, x + e_2\}$. Denote the resulting mark by Y_x. Every unordered edge is assigned once, so the marks $(Y_x)_{x\in\mathbb{Z}^2}$ are iid. Each mark includes its entire future clock histories.

The event A that the eventual configuration is all 1 is measurable in this marked field: eventual states are limits of the measurable integer-time states. It is invariant under all spatial translations, since the graphical construction commutes with them. We recall the elementary zero-one argument for this situation. For any $\epsilon > 0$, choose an event B depending

on finitely many spatial marks with $\mathbb{P}(A \triangle B) < \epsilon$. Such approximations hold for every event in a product sigma field: the class of approximable events contains the finite-coordinate events and is closed under complements and countable unions, the latter by first taking a finite prefix with arbitrarily small probability error. This also covers completed null events.

Choose a spatial translate $\theta_z B$ whose coordinates are disjoint from those of B. The two events are independent, while $\theta_z A = A$. Consequently,

$$|\mathbb{P}(A) - \mathbb{P}(B)^2| \le 2\epsilon, \qquad |\mathbb{P}(A)^2 - \mathbb{P}(B)^2| \le 2\epsilon.$$

It follows that $\mathbb{P}(A)(1 - \mathbb{P}(A)) \le 4\epsilon$. Letting ϵ decrease to zero shows that $\mathbb{P}(A)$ is 0 or 1. Together with Lemma 3, this says that the limiting color is deterministic.

If that color were 2, bounded convergence would give $m(t) \to -1$. But (6) gives $m(t) \ge m(0)$ for every t, and

$$m(0) = a_1 - a_2 = 2a_1 + a_0 - 1 > -1.$$

This contradiction proves that every site is eventually in state 1. For finite nonempty K, the maximum of its finitely many site fixation times is almost surely finite. Its all-CO indicator therefore tends to 1; bounded convergence gives (1). For empty K the probability is identically 1. The bounded-convergence statements apply along every real sequence tending to infinity and hence give the full continuous-time limits. □

A Graphical construction and the counting identity

For completeness, we justify the two clock facts used in the proof. The independent initial states and the exponential interarrival variables of the countably many clocks can be constructed on a countable product of real spaces. Their finite product laws are consistent, so the countable extension theorem applies [3, Section 2.1.4, Theorem 2.1.21]. The exponential construction gives the required independent Poisson histories [3, Section 3.7].

Put $\lambda = q/4 + r/4 + \gamma$, the total edge-clock rate. For a fixed nearest-neighbor path of n steps and a fixed time horizon T, the expected number of its backward, strictly time-ordered edge-ring sequences is $\lambda^n T^n / n!$. This includes paths that repeat an edge. To verify the formula, first count rings in distinct ordered dyadic time bins. Independent counts in disjoint bins give the product of their means. Refinement increases these counts to all strictly ordered tuples, so monotone convergence gives the ordered time simplex of volume $T^n/n!$. There are 4^n spatial paths from a site. Therefore

$$\mathbb{P}(\text{a backward edge-ring path of } n \text{ steps before } T) \le \frac{(4\lambda T)^n}{n!} \longrightarrow 0.$$

To determine a state, trace the most recent incident mark backwards and, at an edge mark, inspect the two predecessor endpoint states. The dependency tree branches at most twice. An infinite branch must make infinitely many spatial steps, because each site has only finitely many incident marks before T. The preceding bound excludes such a branch. A finitely branching infinite tree would contain an infinite branch, so the dependency tree is finite almost surely. Its leaves are initial states, and its updates uniquely determine the state in question. Intersect over sites and integer horizons. The resulting paths are right-continuous with left limits, their changes occur only at incident marks, and they are measurable and nonanticipating. The construction commutes with lattice translations. No infinite-volume independence of the evolved states is required.

To prove (2), partition $[0, T]$ dyadically and on $(t_k, t_{k+1}]$ replace $h(\eta_{t-}|_F)$ by $h(\eta_{t_k}|_F)$. This coefficient is measurable at time t_k; the future increment of the clock is independent of all past data and has mean $c(t_{k+1} - t_k)$. Thus the identity holds for each step approximation.

There are only finitely many marks incident to F before T, and almost surely none at a dyadic endpoint. The approximations therefore converge to the left-state value at every ring of C and at almost every time. If $|h| \le M$, their clock sums are bounded by $MC(0,T]$, of expectation McT, and their time integrands are bounded by M. Dominated convergence proves (2), for signed h as well. Values at fixed deterministic times agree almost surely with their left limits.

AI use

OpenAI Codex was used to develop and check the argument and to draft this note. These checks are AI based; independent mathematical review is requested.

References

[1] R. Durrett and G. Swindle, *Coexistence results for catalysts*, Probability Theory and Related Fields **98** (1994), 489–515. doi:10.1007/BF01192836.

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